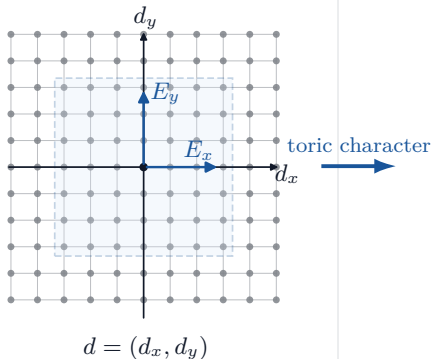


1. Relative lattice

Relative lattice $\mathbb{Z}^2$



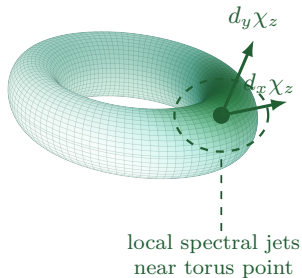
$B|_{\Omega_{\text{vis}}}$ observed relative table
 $\chi_z(d) = z_x^{d_x} z_y^{d_y} = e^{i(\omega_x d_x + \omega_y d_y)}$

2. Spectral torus and jets

Spectral torus / Fourier chart

$$(S^1)^2 \subset (\mathbb{C}^*)^2$$

$$z = (z_x, z_y) = (e^{i\omega_x}, e^{i\omega_y})$$



Spectral flow

$$\partial_\tau e^{i\omega(\tau) \cdot d} \Big|_0 = i(\dot{\omega} \cdot d) e^{i\omega \cdot d}$$

3. Toric jets and PJ modes

Toric jet space

J0 layer: χ_z
 order-1 layer: $d_x \chi_z, d_y \chi_z$
 order-2+ layers: $d_x^a d_y^b \chi_z$
 $a, b \in \mathbb{N}_0, 2 \leq a + b \leq R$

lower-cost
directional
subfamily

Directional PJ modes

$$\phi_{\alpha, u, j}(d) = (u^\top d)^j e^{i\omega_\alpha^\top d}$$

$$0 \leq j \leq R, \alpha \text{ indexes frequencies,}$$

$$u \in \mathbb{R}^2 \setminus \{0\}$$